LECTURE 7

Ex

$$\left\{\begin{array}{rcrcrcr} x_1 & - & x_2 & + & x_3 & = & 3 \\ & & x_2 & - & 2x_3 & = & 1 \\ x_1 & & & + & x_3 & = & 4 \end{array}\right\} \quad \begin{array}{l} (e_1) \\ (e_2) \\ (e_3) \end{array}$$

$$\left\{\begin{array}{rcrcrcr} x_1 & - & x_2 & + & x_3 & = & 3 \\ & & x_2 & - & 2x_3 & = & 1 \\ & & x_2 & & & = & 1 \end{array}\right\} \quad \begin{array}{l} (e_1) \\ (e_2) \\ (e_3) \end{array} \longleftarrow (e_3) + (-1)(e_1)$$

$$\left\{\begin{array}{rcrcrcr} x_1 & - & x_2 & + & x_3 & = & 3 \\ & & x_2 & & & = & 1 \\ & & x_2 & - & 2x_3 & = & 1 \end{array}\right\} \quad \begin{array}{l} (e_1) \\ (e_2) \longleftrightarrow (e_3) \\ (e_3) \longleftrightarrow (e_2) \end{array}$$

$$\left\{\begin{array}{rcrcrcr} x_1 & - & x_2 & + & x_3 & = & 3 \\ & & x_2 & & & = & 1 \\ & & & - & 2x_3 & = & 0 \end{array}\right\} \quad (e_3) \longleftarrow (e_3) + (-1)(e_2)$$

Thus the sol'n set is $\left\{\begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}\right\}$.

Note

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 1 & -2 & 1 \\ 1 & 0 & 1 & 4 \end{array}\right] \Rightarrow 3 \times 4 \text{ matrix}$$

can use R_1, R_2, R_3 instead

Ex

$$\left\{\begin{array}{rcrcrcr} x_1 & + & x_2 & + & 3x_3 & = & 3 \\ & & x_2 & + & x_3 & = & 2 \\ x_1 & + & 2x_2 & + & 4x_3 & = & 2 \end{array}\right\} \quad \begin{array}{l} (e_1) \\ (e_2) \\ (e_3) \end{array}$$

$$\left\{\begin{array}{rcrcrcr} x_1 & + & x_2 & + & 3x_3 & = & 3 \\ & & x_2 & + & x_3 & = & 2 \\ & & x_2 & + & x_3 & = & -1 \end{array}\right\} \quad \begin{array}{l} (e_1) \\ (e_2) \\ (e_3) \end{array} \longleftarrow (e_3) + (-1)(e_1)$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 2 \end{array}\right] \xrightarrow{R_3 + (-1)R_1} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & -1 \end{array}\right] \xrightarrow{R_3 + (-1)R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

Since $0 \neq -3$, there is no sol'n.

CONTINUED

Def'n A <u>$m \times n$ matrix with entries</u> in $\mathbb{F}$ is a rectangular array consisting of m rows and n columns. The entries of the array are numbers in $\mathbb{F}$

Ex

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \Rightarrow 2 \times 2$$

$$B = \begin{bmatrix} 1 & i & 3-i \\ 5 & 0 & 0 \end{bmatrix} \quad 2 \times 3 \text{ w/ entries in } \mathbb{C}$$

Notation We write

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]$$

We write $M_{m \times n}(\mathbb{F})$ for the <u>set</u> of all $m \times n$ matrices w/ entries in $\mathbb{F}$.

Ex

$$M_{2\times 2}(\mathbb{R}) = \left\{ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} : a_{ij} \in \mathbb{R} \right\}$$

$$M_{3\times 1}(\mathbb{C}) = \left\{ \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} : a_{ij} \in \mathbb{C} \right\} = \mathbb{C}^3$$

Ex Solve $\left\{ \begin{array}{l} x_1 + x_3 = 1 \\ 2x_1 - x_2 - 2x_3 = 1 \end{array} \right\}$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & -1 & -2 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -1 & -4 & -1 \end{array}\right] \xrightarrow{(-1)R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 4 & 1 \end{array}\right]$$

So $x_1 + x_3 = 1 \Rightarrow x_1 = 1 - x_3$

$x_2 + 4x_3 = 1 \Rightarrow x_2 = 1 - 4x_3$

We have

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1-t \\ 1-4t \\ 0+t \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix}$$

$m \times n$ matrix w/ entries in $\mathbb{F}$